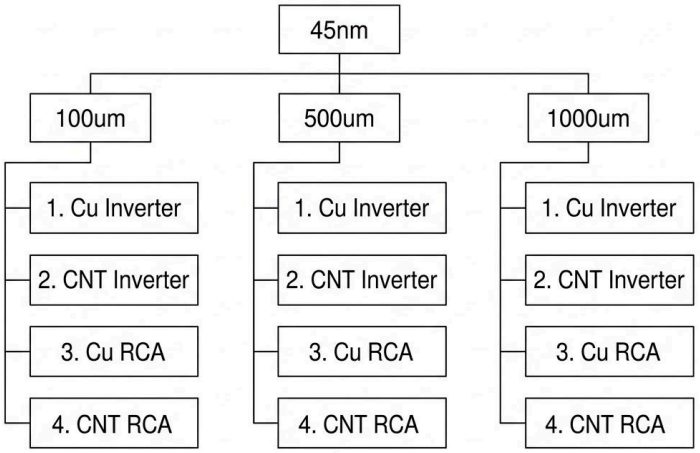
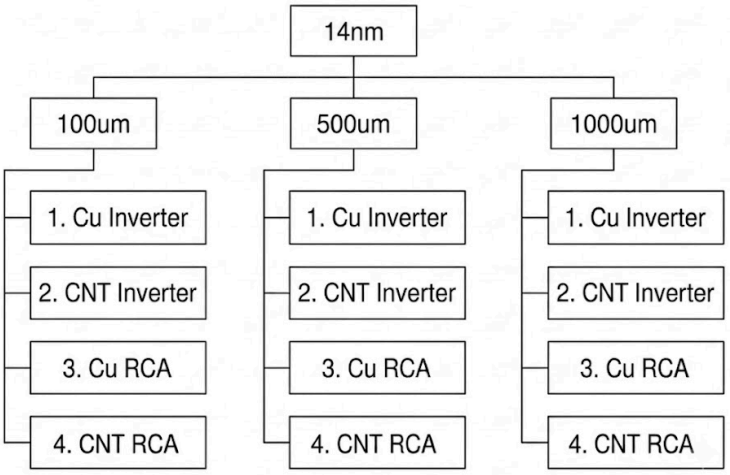
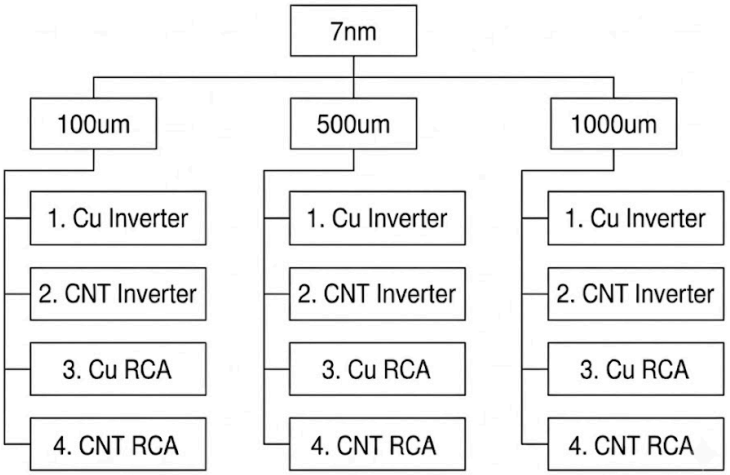
45nm Hierarchy



14nm Hierarchy



7nm Hierarchy



1. Cu Inverter

1. **Run** the simulation.
2. **Right-click on the black waveform area.**
3. Select **Add Traces...**
4. Add:
    - `V(in)`
    - `V(mid)`
    - `V(in2)`
    - `V(out)`
5. Click **OK**.

---

2. CNT Inverter

1. **Run** the simulation.
2. **Right-click on the black waveform area.**
3. Select **Add Traces...**
4. Add:
    - `V(in)`
    - `V(mid)`
    - `V(in2)`
    - `V(out)`
5. Click **OK**.

---

3. Cu 4-bit RCA

1. **Run** the simulation.
2. **Right-click on the waveform area.**
3. Select **Add Traces...**
4. Add:
    - `V(n_c1_raw)`
    - `V(c1)`
5. Click **OK**.

Then create the second plot:

6. **Right-click on the waveform area again.**
7. Select **Add Plot Pane Below**.
8. In the new bottom pane, **right-click**.
9. Select **Add Traces...**
10. Add:
    - `V(s0)`
    - `V(s1)`
    - `V(s2)`
    - `V(s3)`
    - `V(cout)`

11. Click **OK**.
12. Go to **View → SPICE Error Log**.
13. Find `tpd_int1` and record the value.

---

4. CNT 4-bit RCA

1. **Run** the simulation.
2. **Right-click on the waveform area.**
3. Select **Add Traces...**
4. Add:
    - `V(n_c1_raw)`
    - `V(c1)`
5. Click **OK**.

Then:

6. **Right-click on the waveform area again.**
7. Select **Add Plot Pane Below**.
8. In the new bottom pane, **right-click**.
9. Select **Add Traces...**
10. Add:
    - `V(s0)`
    - `V(s1)`
    - `V(s2)`
    - `V(s3)`
    - `V(cout)`
11. Click **OK**.
12. Go to **View → SPICE Error Log**.
13. Find `tpd_int1` and record the value.